

Cybersecurity & Ethical Hacking Roadmap

A Practical Step-by-Step Journey for Building Real Skills, Certifications, and Career Growth

This roadmap is designed for students who want to build a strong career in cybersecurity, ethical hacking, red teaming, penetration testing, and cyber defense. The goal is not only to collect certificates, but to become truly skilled through: Strong technical foundations Daily practice Hands-on labs Real-world problem solving Consistency and patience

Stage 1 — Networking Foundations (1–2 Months)

- Learn IP addressing, subnetting, DNS, HTTP/HTTPS
- Understand TCP/IP and the OSI Model
- Learn ports and protocols
- Study routers, switches, VPNs, and firewalls
- Practice packet analysis basics

Recommended Platforms: Cisco Networking Academy, TryHackMe Pre-Security

Stage 2 — Linux & Operating Systems (1–2 Months)

- Install Kali Linux or Ubuntu in VirtualBox
- Learn terminal commands and Bash basics
- Understand file permissions and users/groups
- Learn networking commands and process management
- Practice daily Linux usage

Goal: Become comfortable using Linux without fear.

Stage 3 — Programming Basics with Python (1–2 Months)

- Variables, loops, functions
- File handling and automation
- Working with APIs and requests
- Simple cybersecurity scripts
- Basic debugging and logic building

Goal: Understand scripts and automate tasks.

Stage 4 — Cybersecurity Fundamentals (2–3 Months)

- CIA Triad
- Authentication and authorization
- Malware and attack types
- Encryption and hashing
- SIEM and SOC basics
- Security policies and vulnerabilities

Recommended Certifications: Security+ or Google Cybersecurity Certificate

Stage 5 — Ethical Hacking Basics (4–6 Months)

- Nmap and reconnaissance
- Burp Suite and web testing
- Wireshark packet analysis
- Metasploit basics
- Web vulnerabilities and OWASP Top 10
- Privilege escalation basics

Recommended Platforms: TryHackMe, Hack The Box, PortSwigger Academy

Stage 6 — Advanced Pentesting & Red Teaming (1–2 Years)

- Active Directory attacks
- Pivoting and lateral movement
- Advanced privilege escalation
- Cloud security basics
- Red teaming methodologies
- AV evasion and advanced labs

Recommended Certifications: PNPT, CPENT, OSCP

Recommended Timeline

Time Period	Focus Area
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Months 1–2	Networking Fundamentals
Months 3–4	Linux & Virtual Machines
Months 5–6	Python & Cybersecurity Basics
Months 7–12	Ethical Hacking Labs & Tools
Year 2	Advanced Pentesting & Certifications
Year 3+	Specialization & Real Industry Experience

Mindset for Success

Cybersecurity is not mastered in a few weeks. The strongest professionals build skills through: Daily consistency
Hands-on labs Curiosity and research Patience during difficult topics Continuous learning Remember: **Real skill matters more than collecting certificates.**

Final Advice

Do not compare your progress with others. Focus on becoming better every month. Build a strong foundation first, then slowly move toward advanced ethical hacking, penetration testing, and red teaming. With patience, discipline, and practice, you can build a strong cybersecurity career.